

Final Project Documentation

Support Agent Performance Analysis (MySQL and Tableau)

Sessions documented: 30 September to 7 October 2026 | Prepared for: Aaron

One-line summary: Ranking support agents by overall resolution time is misleading: in this simulated dataset, ticket priority mix accounts for most of the gap between the fastest and slowest agent (†), while channel mix accounts for almost none (†).

Items marked (†) were calculated outside MySQL during review. Reproduce them in SQL or leave them out before publishing. Numbers were checked on the CSV, so confirm them against your own MySQL and Tableau output.

Session Log

Date	Focus	Outcome
30 Sep	Reviewed exploratory queries; fixed vs escalated rates; built the agent scorecard in one CTE	Scorecard with counts, rates, ranks and three averages; MySQL error 1055 fixed
1 Oct	Fairness check by priority; channel and category analysis; channel mix by agent (two CTEs and a JOIN); dataset source review	Channel mix explains almost none of the agent gap; priority mix explains most (†)
2 Oct	Connected Tableau to MySQL; Custom SQL data sources; highlighted rate bars; priority heatmap	First two worksheets
3 Oct	Priority mix, channel mix, profile table, channel and category heatmap, top 10; KPI cards and calculated fields	All charts built and checked against SQL
4 to 5 Oct	Dashboard assembly: containers, KPI row, layout, titles, spacing	Charts placed in five rows; KPI row correct
6 to 7 Oct	KPI sheet with Measure Names and Measure Values, colors, axis names, legends container; MySQL login fix for Tableau	Dashboard finished

1. Project Title

Support Agent Performance Analysis: a fair comparison of support agents using SQL in MySQL and a Tableau dashboard.

2. Problem Statement

Practice scenario. The source dataset does not state a business problem. A customer support team at a multi-product SaaS company handles tickets across several channels, categories and priority levels. Managers want to know how individual agents perform, but each ticket type is different, so a simple comparison can be unfair or misleading. It is unclear whether differences between agents reflect real performance or only the kinds of tickets each agent receives.

3. Objective

- Measure each agent's fixed rate and escalation rate, and flag the highest escalation rate as a watch-list signal.
- Build an agent scorecard: volume, rates, ranks, first response time, resolution time and CSAT.
- Test whether ticket priority, channel or category explains differences in resolution time between agents.
- Present the results in a Tableau dashboard that shows values and ticket counts, not just ranks.

4. Dataset Information

Item	Detail
Source	Simulated customer support system for a multi-product SaaS company (7 channels). Dataset name and link: not recorded, add before publishing.
Size	Source describes 2,000 tickets. After cleaning, table tickets_raw_staging has 1,924 rows and 23 columns, one row per ticket. The original table tickets_raw also has 1,924 rows, so the cleaning script removed none. 76 ticket IDs (of TKT-00001 to TKT-02000) are absent, and TKT-00021 is not in the original downloaded file either, so the gap comes from the source file.
Period	Tickets created from 2024-01-01 to 2025-06-30
Coverage	9 agents (about 194 to 247 tickets each), 7 channels, 7 languages, 5 products, 7 regions, 4 customer plans
Key columns	ticket_id, created_at, resolved_at, channel, category, priority, status, sentiment, product, assigned_agent, agent_id, first_response_time_hrs, resolution_time_hrs, csat_score, escalated, reopened
Data notes	resolution_time_hrs is blank for the 532 Open or In Progress tickets. csat_score is blank for all Escalated tickets and open tickets (644). escalated is stored as text ('True' / 'False') in MySQL. Status values: Open, In Progress, Resolved, Closed, Escalated.

Data cleaning performed

Data problem	Why it matters	Cleaning approach
Blank text instead of NULL (resolution note, resolution time, CSAT, first response time, reopened, escalated, interactions, tenure)	Blanks are not NULL, so AVG() and COUNT(column) treat them as values	UPDATE ... SET column = NULL WHERE column = "
Dates stored as text	Dates cannot be sorted or calculated as text	Blank to NULL, STR_TO_DATE, then ALTER to DATETIME
Extra spaces in text columns	'Email ' and 'Email' would be counted as different values	TRIM() on all text columns
Blank agent_id values	Agents could not be grouped reliably	Filled from assigned_agent with REGEXP_SUBSTR(assigned_agent, '[0-9]+')
Free-text message column	Not used in this analysis	ALTER TABLE ... DROP COLUMN message

Validation: the script works on a staging copy (tickets_raw_staging, created from tickets_raw). Row counts are the same before and after cleaning (1,924 in both tables), so the cleaning script removed no rows. The 76 missing ticket IDs were already absent from tickets_raw and from the original downloaded file (checked for TKT-00021). The cleaned data has 1,924 unique ticket IDs.

5. Problems Identified

- **Source versus file:** 2,000 tickets described, 1,924 present (76 ticket IDs absent from the source file); 5 sentiment levels described, 4 present (Very Negative, Negative, Neutral, Positive); "6 languages" described but 7 listed and present.
- **Blank values:** resolution time (532) and CSAT (644), caused by open and escalated tickets.
- **Early queries:** "fixed" counted every non-escalated ticket, including 532 still open; the product and category questions grouped by two columns.
- **Misleading overall ranking:** resolution time depends heavily on priority (about 7 to 10 hrs for Critical against 99 to 111 hrs for Low), and agents receive different priority mixes.
- **Small groups:** as few as 4 Critical tickets for one agent, 7 to 17 escalations per agent, and channel and category combinations with 10 or fewer tickets.
- **Tableau data problems:** an average of category averages gave a wrong channel chart; counts instead of shares gave unequal bars; a copied calculated field became invalid; percentages multiplied by 100 twice.

- **Login:** Tableau could not sign in to MySQL 8 after a restart (caching_sha2_password needs a secure first connection).

6. Challenges Encountered

- **Understanding results:** expecting the agent with the best fixed rate (Omar F) to also be fastest and have the best CSAT; the metrics measure different things.
- **Writing queries:** aliasing COUNT(*) as a column name, grouping by the wrong columns, running a CTE alone, and the syntax for two CTEs and a JOIN.
- **NULLs:** COUNT(*) versus COUNT(column) gave different ticket counts next to an average.
- **Sorting:** a second ORDER BY key of agent_id meant averages were not ascending within each priority.
- **SQL errors:** error 1055 (ONLY_FULL_GROUP_BY) and error 1054 (a misspelled column name).
- **Tableau:** connection settings, a relationship alert from two tables on one canvas, Compute Using for table calculations, Measure Names and Measure Values, Floating versus Tiled containers, and spacing, legends and cut-off text on the dashboard.

7. SQL and Tableau Techniques Used

Technique	Where it was used
SELECT, aliases, ROUND, AVG, COUNT	All queries
COUNT(column) versus COUNT(*)	Count of tickets behind each average
GROUP BY (one to three columns), HAVING	By agent; agent and priority; channel and category; hide groups of fewer than 10 tickets
CASE inside COUNT	Fixed, still open and escalated counts
CTEs (two in one query) and JOIN	Share of an agent's tickets by channel and by priority
Window function RANK() OVER	Agent ranks in fixed rate and escalation rate
NULLIF, ORDER BY (multi-level), FIELD()	Avoid division by zero; custom priority order
Custom SQL data sources	One Tableau data source per query
Calculated fields and table calculations	Rates, weighted averages, highlights (WINDOW_MAX), heatmap color (WINDOW_AVG), Compute Using
Measure Names and Measure Values, Top N filter	KPI sheet and profile table; top 10 chart
Dashboard containers, Fit, padding, legends	Dashboard layout

8. Queries Used

Workbench versions. For Tableau Custom SQL, the same queries were pasted without the final ORDER BY, comments and semicolon.

Data cleaning script (condensed; each UPDATE is shown for one or two columns, the script covers all)

```
CREATE TABLE tickets_raw_staging LIKE tickets_raw;
INSERT INTO tickets_raw_staging SELECT * FROM tickets_raw;

-- duplicate check (SELECT only)
SELECT ticket_id, COUNT(*) AS copies
FROM tickets_raw_staging GROUP BY ticket_id HAVING COUNT(*) > 1;

-- blanks to NULL, then convert dates
UPDATE tickets_raw_staging SET created_at = NULL WHERE created_at = '';
UPDATE tickets_raw_staging SET resolved_at = NULL WHERE resolved_at = '';
UPDATE tickets_raw_staging
  SET created_at = STR_TO_DATE(created_at, '%Y-%m-%d %H:%i:%s'),
      resolved_at = STR_TO_DATE(resolved_at, '%Y-%m-%d %H:%i:%s');
ALTER TABLE tickets_raw_staging MODIFY COLUMN created_at DATETIME;
ALTER TABLE tickets_raw_staging MODIFY COLUMN resolved_at DATETIME;

-- trim spaces, other blanks to NULL
UPDATE tickets_raw_staging SET channel = TRIM(channel), category = TRIM(category);
UPDATE tickets_raw_staging SET csat_score = NULL WHERE csat_score = '';

-- drop the unused column, fill blank agent IDs
ALTER TABLE tickets_raw_staging DROP COLUMN message;
UPDATE tickets_raw_staging SET agent_id = REGEXP_SUBSTR(assigned_agent, '[0-9]+')
WHERE agent_id IS NULL OR agent_id = '';
```

Query 1: Agent scorecard

```
WITH agent_fixed_escalation AS
(
  SELECT agent_id, assigned_agent,
    COUNT(CASE WHEN escalated = 'False'
      AND status IN ('Resolved','Closed') THEN 1 END) AS total_fixed,
    COUNT(CASE WHEN escalated = 'False'
      AND status IN ('Open','In Progress') THEN 1 END) AS total_still_open,
    COUNT(CASE WHEN escalated = 'True' THEN 1 END) AS total_escalated,
    COUNT(*) AS total_agent_handled,
    AVG(first_response_time_hrs) AS avg_frh,
    AVG(resolution_time_hrs) AS avg_rth,
    AVG(csat_score) AS avg_csatscore
  FROM tickets_raw_staging
  GROUP BY agent_id, assigned_agent
)
SELECT agent_id, assigned_agent, total_fixed, total_still_open, total_escalated,
  total_agent_handled,
  ROUND(100.0 * total_fixed / NULLIF(total_fixed + total_escalated, 0), 1) AS fixed_rate_pct,
  ROUND(100.0 * total_escalated / NULLIF(total_fixed + total_escalated, 0), 1) AS escalated_rate_pct,
  RANK() OVER (ORDER BY 1.0 * total_fixed
    / NULLIF(total_fixed + total_escalated, 0) DESC) AS agent_rank_in_fixed,
  RANK() OVER (ORDER BY 1.0 * total_escalated
    / NULLIF(total_fixed + total_escalated, 0) DESC) AS agent_rank_in_escalation,
  ROUND(avg_frh, 1) AS avg_response_time,
  ROUND(avg_rth, 1) AS avg_resolution_time,
  ROUND(avg_csatscore, 1) AS avg_csat_score
FROM agent_fixed_escalation
ORDER BY agent_id ASC;
```

Query 2: Average resolution time by agent and priority

```
SELECT agent_id, assigned_agent, priority,
  COUNT(resolution_time_hrs) AS tickets_counted,
  ROUND(AVG(resolution_time_hrs), 1) AS avg_resolution_time
FROM tickets_raw_staging
GROUP BY agent_id, assigned_agent, priority
ORDER BY FIELD(priority, 'Critical', 'High', 'Medium', 'Low'),
  avg_resolution_time ASC,
  agent_id;
```

Query 3: Resolution time by channel (reconstructed from the session output)

```
SELECT channel,
  COUNT(resolution_time_hrs) AS channel_counted,
  ROUND(AVG(resolution_time_hrs), 1) AS avg_resolution_time
FROM tickets_raw_staging
GROUP BY channel
ORDER BY avg_resolution_time DESC;
```

Query 4: Resolution time by channel and category

```
SELECT channel, category,
       COUNT(resolution_time_hrs) AS tickets_counted,
       ROUND(AVG(resolution_time_hrs), 1) AS avg_resolution_time
FROM tickets_raw_staging
GROUP BY channel, category
HAVING tickets_counted >= 10
ORDER BY avg_resolution_time DESC;
```

Query 5: Channel share by agent (two CTEs and a JOIN)

```
WITH first_cte AS
(
  SELECT agent_id, assigned_agent, channel, COUNT(*) AS channel_count_per_agent
  FROM tickets_raw_staging
  GROUP BY agent_id, assigned_agent, channel
),
second_cte AS
(
  SELECT agent_id, COUNT(*) AS total_agent_tickets
  FROM tickets_raw_staging
  GROUP BY agent_id
)
SELECT f.agent_id, f.assigned_agent, f.channel, f.channel_count_per_agent,
       s.total_agent_tickets,
       ROUND(100.0 * f.channel_count_per_agent / s.total_agent_tickets, 1) AS channel_pct
FROM first_cte AS f
JOIN second_cte AS s ON f.agent_id = s.agent_id
ORDER BY f.agent_id, channel_pct DESC;
```

Query 6: Priority share by agent (the same pattern with priority in place of channel; 36 rows)

```
WITH first_cte AS
(
  SELECT agent_id, assigned_agent, priority, COUNT(*) AS priority_count_per_agent
  FROM tickets_raw_staging
  GROUP BY agent_id, assigned_agent, priority
),
second_cte AS
(
  SELECT agent_id, COUNT(*) AS total_agent_tickets
  FROM tickets_raw_staging
  GROUP BY agent_id
)
SELECT f.agent_id, f.assigned_agent, f.priority, f.priority_count_per_agent,
       s.total_agent_tickets,
       ROUND(100.0 * f.priority_count_per_agent / s.total_agent_tickets, 1) AS priority_pct
FROM first_cte AS f
JOIN second_cte AS s ON f.agent_id = s.agent_id
ORDER BY f.agent_id, FIELD(f.priority, 'Critical', 'High', 'Medium', 'Low');
```

Tableau calculated fields

Field	Formula	Use
Is Highest (and the escalation and CSAT versions)	$\text{AVG}([\text{escalated_rate_pct}]) = \text{WINDOW_MAX}(\text{AVG}([\text{escalated_rate_pct}]))$	Marks the highest bar. One field per measure; Compute Using assigned_agent
Diff vs priority average	$\text{SUM}([\text{avg_resolution_time}]) - \text{WINDOW_AVG}(\text{SUM}([\text{avg_resolution_time}]))$	Heatmap color: hours above or below the priority average
channel - category	$[\text{channel}] + " - " + [\text{category}]$	Label for the top 10 chart
Is Highest Combo	$\text{AVG}([\text{avg_resolution_time}]) = \text{WINDOW_MAX}(\text{AVG}([\text{avg_resolution_time}]))$	Marks the slowest combination
Fixed rate	$\text{SUM}([\text{total_fixed}]) / (\text{SUM}([\text{total_fixed}]) + \text{SUM}([\text{total_escalated}]))$	KPI card, Percentage format
Escalation rate	$\text{SUM}([\text{total_escalated}]) / (\text{SUM}([\text{total_fixed}]) + \text{SUM}([\text{total_escalated}]))$	KPI card, Percentage format
Avg first response (hrs)	$\text{SUM}([\text{avg_response_time}] * [\text{total_agent_handled}]) / \text{SUM}([\text{total_agent_handled}])$	KPI card, weighted by all tickets
Avg resolution time (hrs)	$\text{SUM}([\text{avg_resolution_time}] * ([\text{total_fixed}] + [\text{total_escalated}])) / \text{SUM}([\text{total_fixed}] + [\text{total_escalated}])$	KPI card, weighted by tickets with a resolution time
Avg CSAT	$\text{SUM}([\text{avg_csat_score}] * [\text{total_fixed}]) / \text{SUM}([\text{total_fixed}])$	KPI card, weighted by rated tickets

Only one place may multiply a rate by 100: either the formula or the Percentage format.

9. Explanation of Queries and Charts

Item	What it does and why it was used
Scorecard	One row per agent. Rates use tickets that reached an outcome (fixed + escalated), so agents with more tickets get no advantage and open tickets do not count against anyone. The ORDER BY inside RANK() sets how ranks are assigned.
Priority query	Compares agents on the same kind of ticket. A count of the same column as the AVG shows how many tickets each average rests on; FIELD() gives a custom order.
Channel, and channel and category	Tests whether channel or category explains slower resolution. HAVING hides small groups and works because the count exists only after grouping.
Channel and priority share	The first CTE counts tickets per agent and group; the second counts tickets per agent; the JOIN divides one by the other to get a percentage within each agent.
Dashboard	KPI cards, rate and CSAT bars with highlights, an agent profile table, a priority heatmap, priority and channel mix bars, a channel and category heatmap, and a top 10 chart. A separate Dashboard Guide explains how to read each one.

10. Resources Used

- MySQL 8.0 and MySQL Workbench
- Tableau (workbook Performance_Analysis_Visualization)
- Dataset overview (source page text; name and link to be added)
- Tableau Help text on connecting to MySQL (pasted during a session)
- README structure guide (Git & Portfolio carousel, #codepathindia)
- AI assistance (Claude), including web searches about Tableau and MySQL connections

Add any other resource you actually used. None were assumed.

11. Solutions and Fixes

Problem	Fix
"Fixed" counted open tickets	Count fixed as escalated = 'False' AND status IN ('Resolved','Closed'); count open separately.
Rates compared different bases	Use fixed + escalated as the denominator for both rates.
Error 1055 (ONLY_FULL_GROUP_BY)	Every selected column must be in GROUP BY or inside an aggregate such as AVG().
COUNT(*) did not match the average	Use COUNT(resolution_time_hrs) next to AVG(resolution_time_hrs).
Averages not ascending in each priority	ORDER BY FIELD(priority, ...), then the average, then agent_id as tie-breaker.
Error 1054, unknown column	Correct the misspelled column name (assigned_agen to assigned_agent).
Tableau relationship alert	Keep one Custom SQL table per data source; add each query as a new data source.
Custom SQL errors in Tableau	Remove the final ORDER BY, comments and semicolon.
Highlight on the wrong bars	One calculated field per measure, with Compute Using assigned_agent.
Heatmap colored by priority column	Color by the difference from the priority average, with a diverging palette centered at 0.
Channel chart used an average of averages	Chart the channel and category combinations, or use a channel-only query.
Percentage showed 0 or 804.6%	Apply the Percentage format and do not multiply by 100 in the formula.
Sheets would not drop into a container	The container was Floating; switch to Tiled.
Wasted space and cut-off text	Fit Entire View, fixed row heights, shorter titles, legends moved into their own container.
Tableau could not sign in to MySQL	Log in once through Workbench first, or tick Require SSL and choose ca.pem.

12. Key Learnings

- A rate is only meaningful if you can say what its denominator is.
- The GROUP BY columns set the level of detail; more columns mean smaller, noisier groups.
- AVG() skips NULLs, so show a count of the tickets behind each average.
- A rank shows order, not distance; show the values as well.
- Averages are fair only when groups contain similar items, so compare within priority.
- Never average averages across groups of different size; weight them or calculate at the right level.
- A hypothesis that fails (channel) is still a useful result.
- A calculated field serves one measure and depends on its data source's column names.
- Charts need titles and a legend or caption; use color consistently, with red only for the number to watch.
- The data is simulated, so findings are about technique and this dataset, not real agents.

13. Things to Practice

- Write multi-CTE queries from scratch, including the comma between CTEs and the final SELECT.
- Reproduce the (†) checks in SQL (CSAT against speed, and the expected average from each mix).
- Add a reopened-rate KPI and a monthly trend query from created_at.
- Use dashboard actions (Use as Filter), which work between sheets that share a data source.
- Create a read-only MySQL user for Tableau, and choose Live or Extract before publishing.
- Learn Power BI or another dashboard tool as a next step.

14. Final Outcome

Completed

- Cleaned table analysed in MySQL: scorecard, priority fairness check, channel and category analysis, and channel and priority mix by agent.
- Tableau dashboard with seven KPI cards, highlighted rate bars, an agent profile table, two heatmaps, two mix charts and a top 10 chart.
- Documentation, a README template, a Dashboard Guide and a KPI Guide.

To add before publishing

- Dataset name and link; optionally confirm that the original downloaded file has 1,924 data rows.
- A dashboard screenshot and, if published, the Tableau Public link (switch data sources to Extract first).
- Confirmation that the (†) numbers match your own checks.

Key findings

- **Rates differ modestly:** fixed rate 88.8% to 95.5%, escalation rate 4.5% to 11.2% (Raj P highest, 17 of 152). Overall 92.0% fixed and 8.0% escalated.
- **Metrics do not line up:** Omar F has the best fixed rate (95.5%) but mid-table speed and a CSAT of 2.6. Across 1,280 rated tickets, CSAT has almost no relationship with speed (†).
- **Priority matters:** Daniel N is fastest overall (51.7 hrs) but ranks 2nd, 6th, 2nd and 4th within Critical, High, Medium and Low, helped by 10.8% Critical and 26.4% Low tickets. Yuki T has 4.4% Critical and 31.5% Low.
- **Channel mix explains very little:** email is 24% to 30% and chat 20% to 25% of every agent's tickets. In a rough what-if check (†), priority mix accounts for about 8.8 of the 11.8-hour gap, channel mix about 0.5 and category mix about 0.9.
- **Slowest combinations:** whatsapp / General Inquiry 88.5 hrs (10 tickets), helpdesk_portal / Feature Request 75.3 (13), web_form / Delivery / Shipping 74.6 (16). Delivery / Shipping and Feature Request recur.

Limitations

- Simulated data; small groups; CSAT blank for escalated tickets; unclear meaning of resolution time for escalated tickets; the what-if check is a rough estimate; 76 ticket IDs are missing from the source file.

Mistakes to remember

- Grouping by extra columns, and leaving a bare column in a GROUP BY query.
- Using COUNT(*) beside an average of a column with NULLs.
- Averaging averages, and multiplying a percentage by 100 twice.
- Reusing one calculated field on charts that measure something else.